Week 4

Virtual Machine (VM) and Cloud Computing

From Physical Hardware to On-Demand Infrastructure



DSI102 Introduction to Digital Technology

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Where Are We in DSI102?

Week 4 connects operating systems to networked infrastructure.



Course progression

Physical components and operating systems become the foundation for virtual machines, cloud infrastructure, and later computer networks.

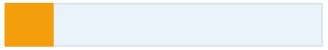
The Problem with Physical Servers

Illustrative example: one server per workload can waste capacity.

Server A

Workload utilization
15%

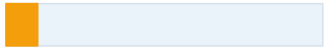
Illustrative example



Server B

Workload utilization
10%

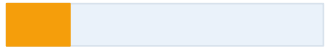
Illustrative example



Server C

Workload utilization
20%

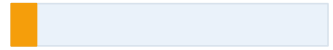
Illustrative example



Server D

Workload utilization
8%

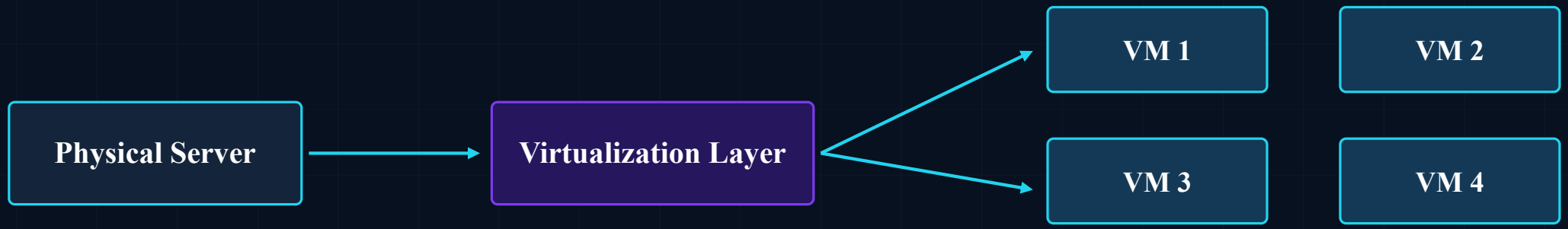
Illustrative example



Why dedicate an entire server to one workload?

One Physical Machine Can Become Many Logical Machines

Virtualization separates what software sees from the underlying physical hardware.



Learning Outcomes

By the end of Week 4, students should be able to explain how virtual infrastructure becomes cloud services.

Virtualization

Explain abstraction of computing resources

Virtual Machines

Describe VM components and behavior

Hypervisors

Distinguish Type 1 and Type 2

Cloud Basics

Relate virtualization to cloud delivery

Service Models

Compare IaaS, PaaS, and SaaS

Deployment & Risk

Identify models, benefits, and security concerns

01

Virtualization

Separating computing from physical hardware



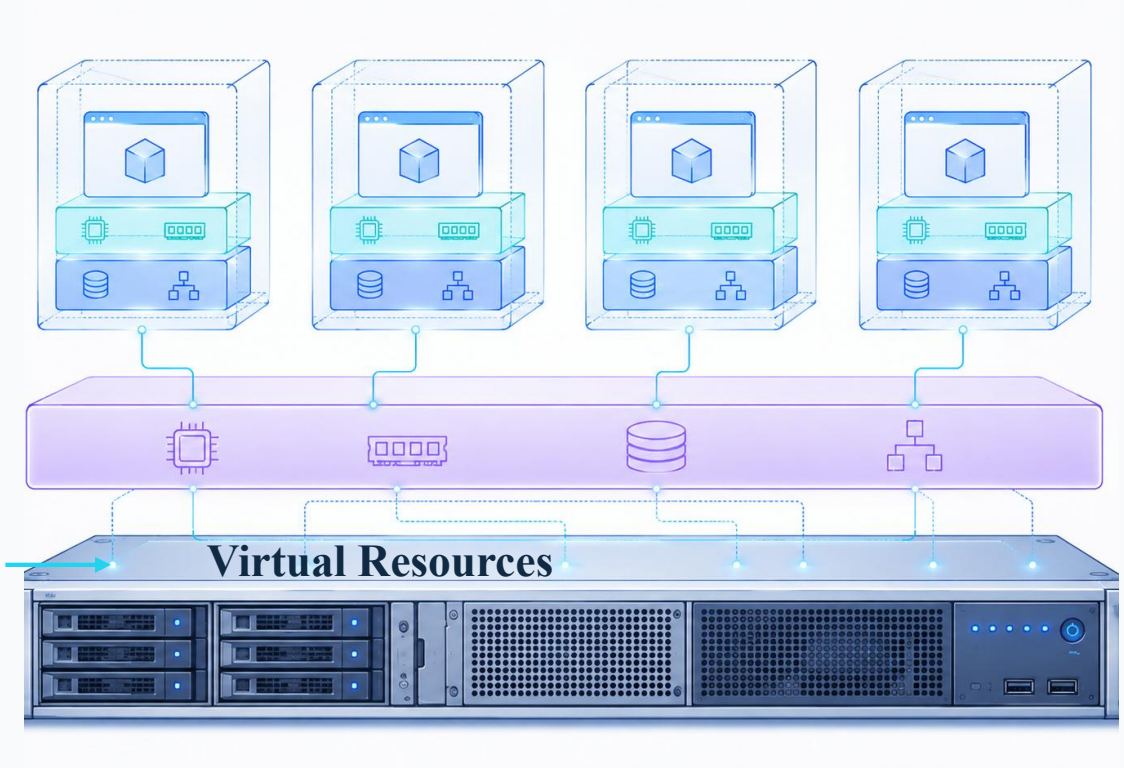
What Is Virtualization?

Virtualization creates logical versions of computing resources.

Definition

Virtualization creates virtual representations of servers, operating systems, storage, or networks.

Physical Resource → **Abstraction Layer** → **Virtual Resources**



Before and After Virtualization

Virtualization consolidates multiple workloads onto shared physical infrastructure.

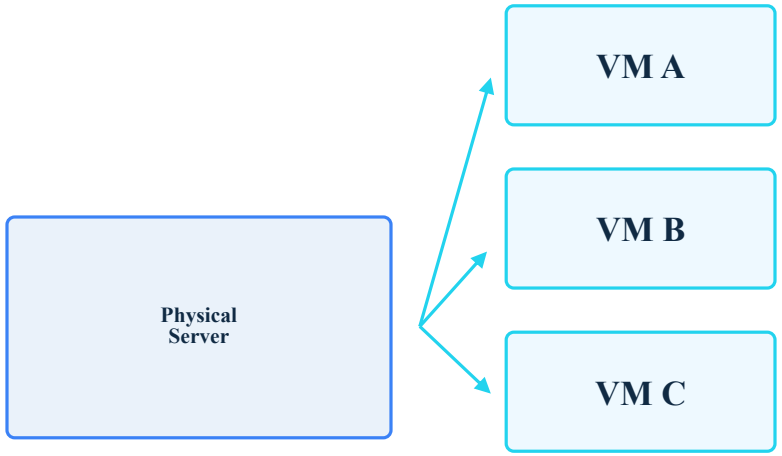
Traditional Infrastructure

One physical server is commonly dedicated to one major workload.



Virtualized Infrastructure

One physical server can host multiple isolated virtual environments.



Physical Machine vs Virtual Machine

A VM behaves like a computer but is defined and managed in software.

Physical Machine

- Dedicated hardware
- Usually runs one primary OS
- Hardware-dependent
- Physical maintenance required
- Direct device ownership

Virtual Machine

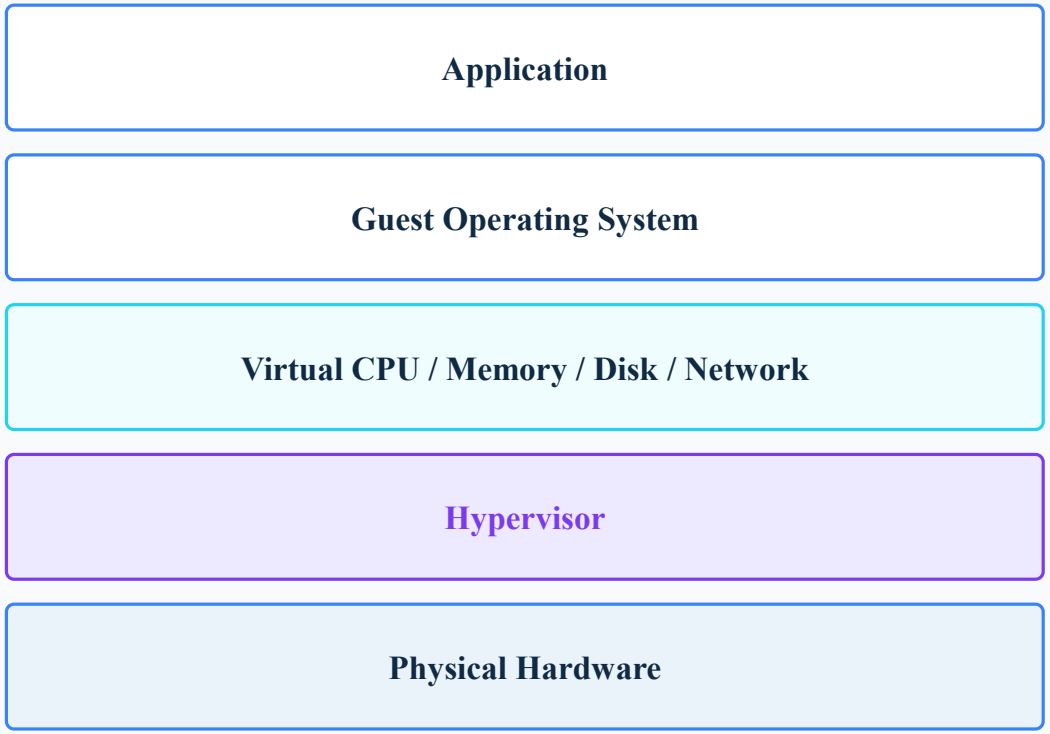
- Software-defined computer
- Shares physical hardware
- Runs its own guest OS
- Portable as files/images
- Isolated operating environment

Anatomy of a Virtual Machine

A VM combines guest software with virtualized hardware resources.

Key idea

The guest OS believes it has hardware, but the hypervisor maps those requests to real physical resources.



Vocabulary

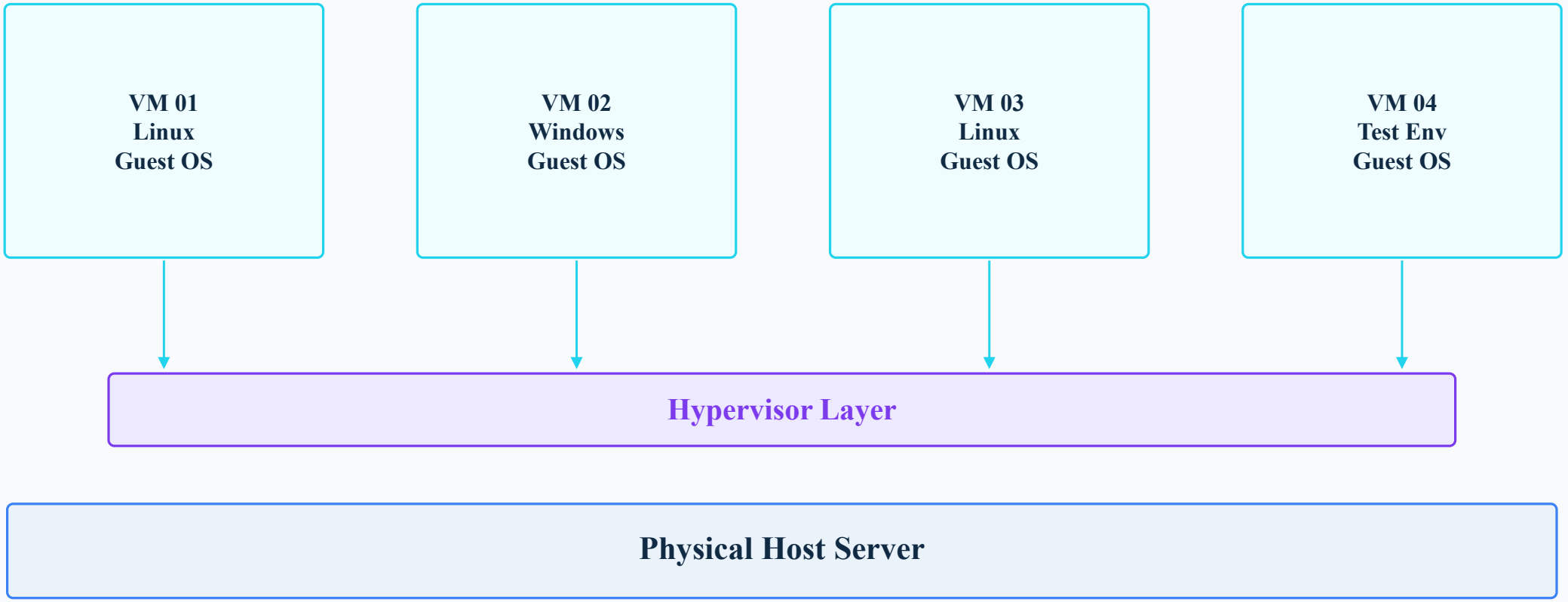
Host = physical or primary environment

Guest = OS running inside the VM

One Host, Multiple Guests

Several isolated operating environments can share one physical server.

Virtual environments are isolated, but they still compete for shared CPU, memory, storage, and network resources.



What Is Virtualized?

A VM sees virtual resources that map onto real physical resources.

CPU

Virtual processors

Mapped by the hypervisor to real host resources.

Memory

Virtual RAM

Mapped by the hypervisor to real host resources.

Storage

Virtual disk

Mapped by the hypervisor to real host resources.

Network

Virtual NIC

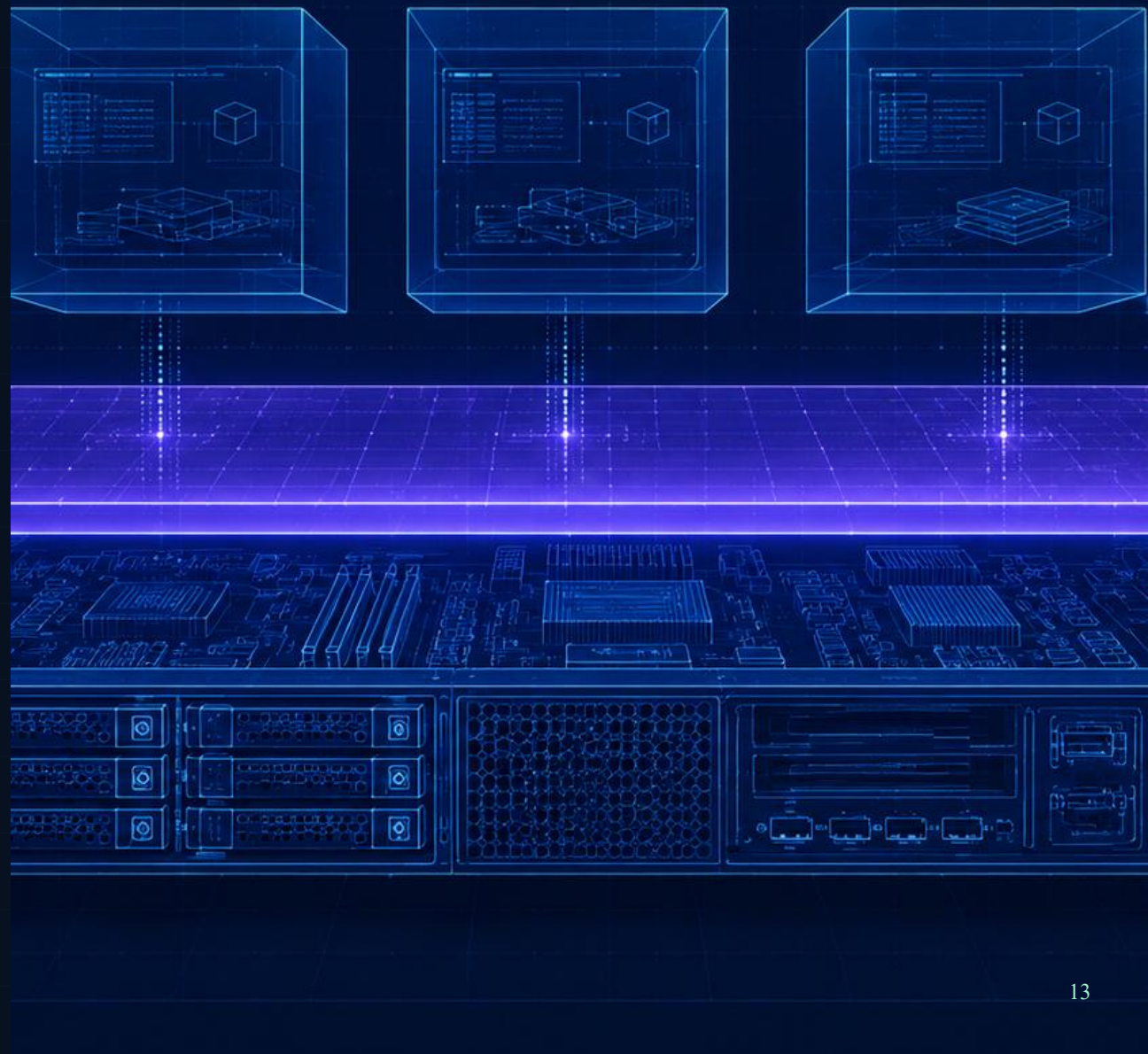
Mapped by the hypervisor to real host resources.

The virtualization layer translates between guest expectations and host reality.

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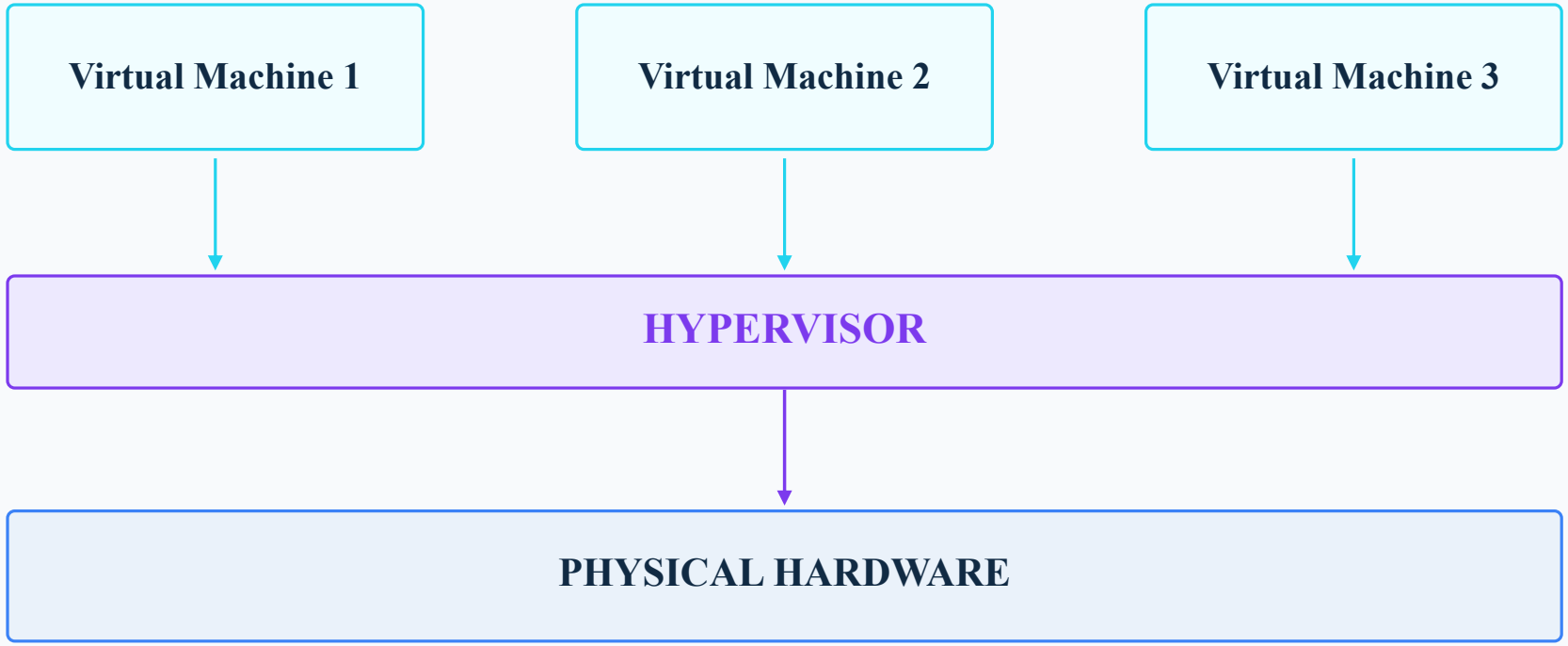
Hypervisors

The control layer that creates and manages virtual machines



What Is a Hypervisor?

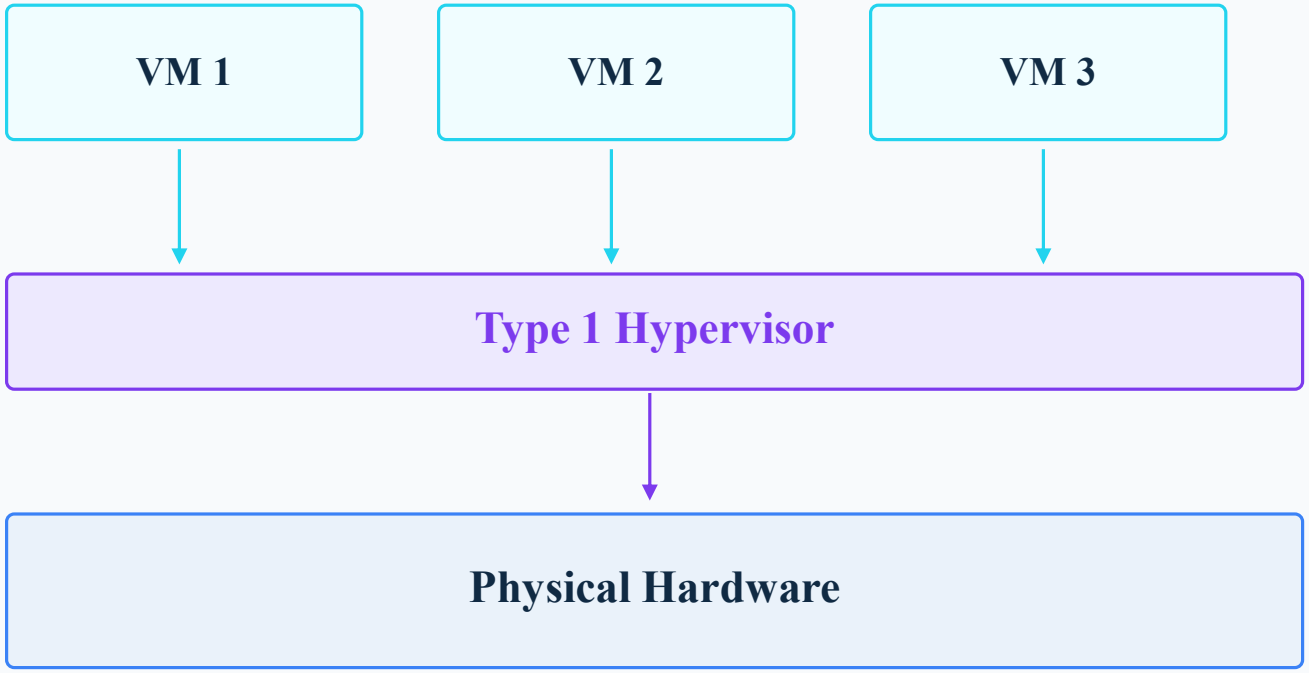
The hypervisor is the software layer that creates and manages VMs.



VMs do not “touch” hardware directly. They interact through the hypervisor.

Type 1 Hypervisor: Bare-Metal

Runs directly on physical hardware, common in data centers and enterprise virtualization.

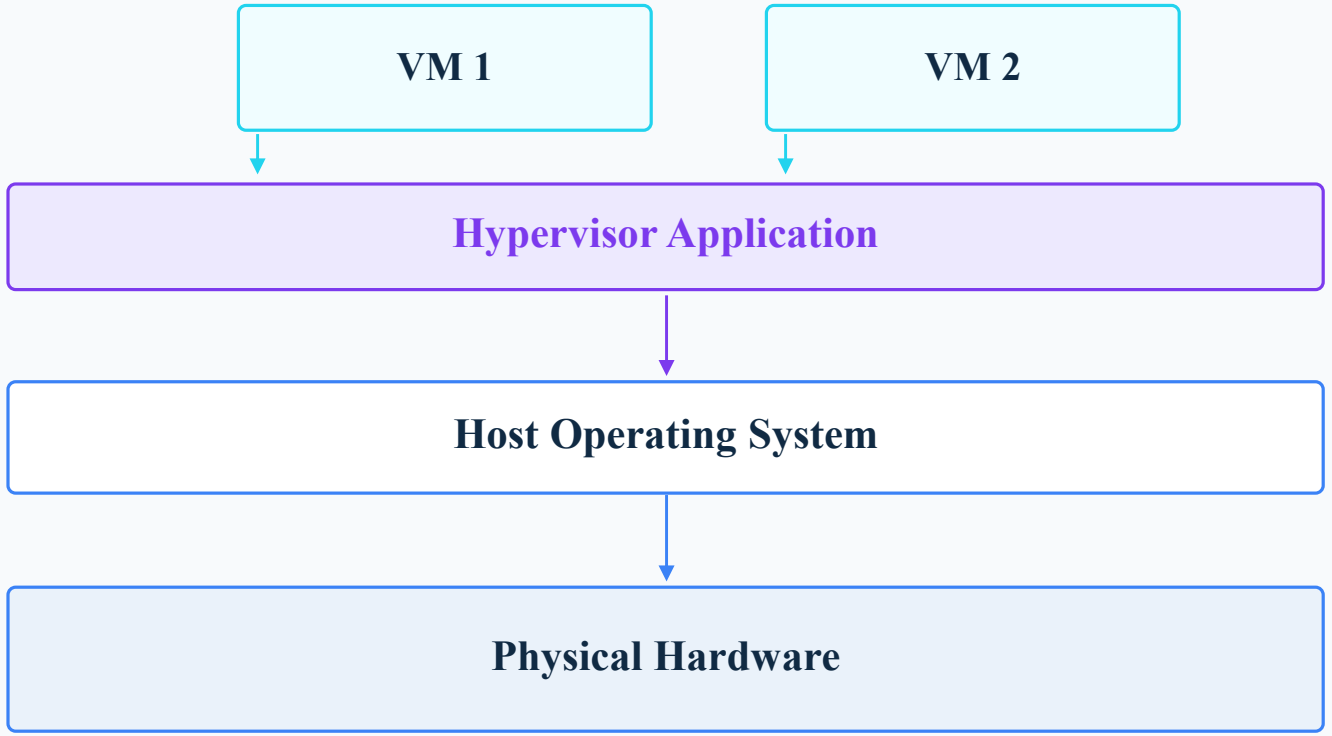


Examples

Examples include VMware ESXi, Hyper-V Server, Xen, and KVM-based environments. These are examples, not endorsements.

Type 2 Hypervisor: Hosted

Runs as an application on top of a host operating system, common for learning and testing.



Examples

VirtualBox, VMware Workstation, and Parallels Desktop are common Type 2 examples.

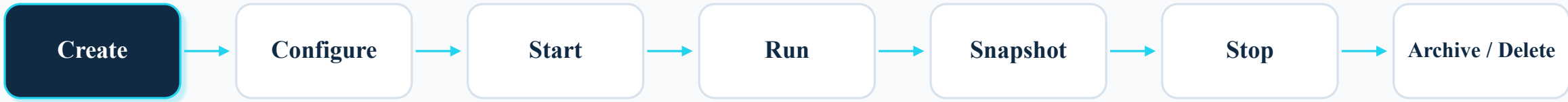
Type 1 vs Type 2 Hypervisors

The main difference is where the hypervisor sits in the architecture.

Criterion	Type 1	Type 2
Position	Directly on hardware	On top of host OS
Host OS required?	No	Yes
Typical environment	Data center / enterprise	Desktop / laptop
Performance	Higher potential performance	More overhead
Common use	Production virtualization	Learning, testing, development

VM Lifecycle

Virtual machines can be created, configured, controlled, saved, and removed as software-defined environments.



VMs are managed objects. They can be copied, paused, restored, and removed more flexibly than physical computers.

Resource Allocation

Physical resources are divided among virtual machines.

Physical Server

CPU: 16 cores
RAM: 64 GB
Storage: 2 TB
Network: 10 Gbps

Illustrative Configuration

Allocated to virtual machines

VM A

4 cores
16 GB RAM

VM B

6 cores
24 GB RAM

VM C

2 cores
8 GB RAM

VM D

4 cores
16 GB RAM

Resource contention can occur when too many VMs compete for the same host capacity.

Snapshots and Cloning

Snapshots preserve a VM state; clones create another VM based on an existing one.



Snapshot

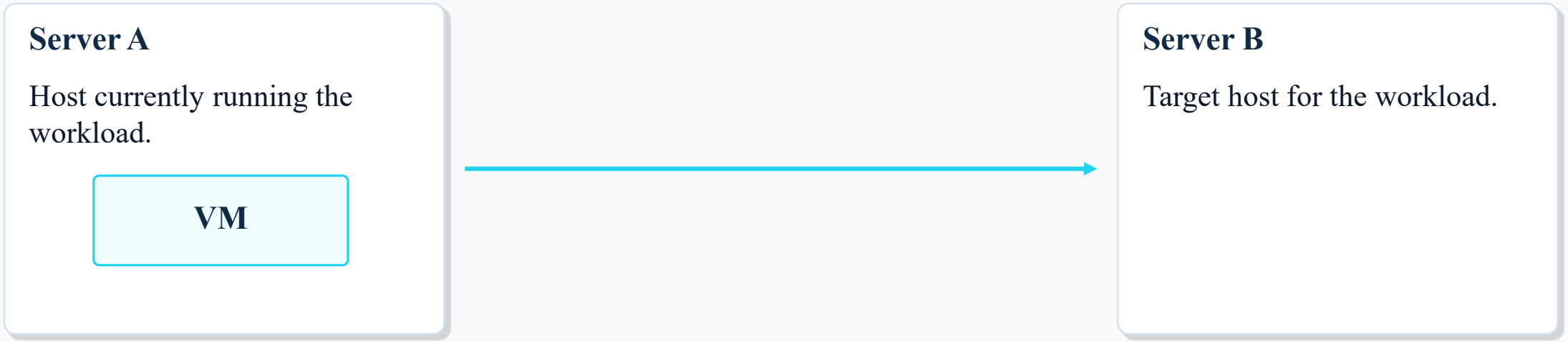
Captures the state of a VM at a point in time. Useful before updates or testing.

Clone

Creates another VM based on an existing VM. Useful for repeated environments.

VM Migration

A VM can move from one physical server to another for maintenance or workload balancing.



Migration supports maintenance, balancing, and availability.

Benefits and Limitations of Virtualization

Virtualization improves flexibility but introduces management and resource trade-offs.

Benefits

- Better resource utilization
- Isolation between environments
- Faster deployment
- Easier testing
- Reduced physical infrastructure
- Portability of VM images

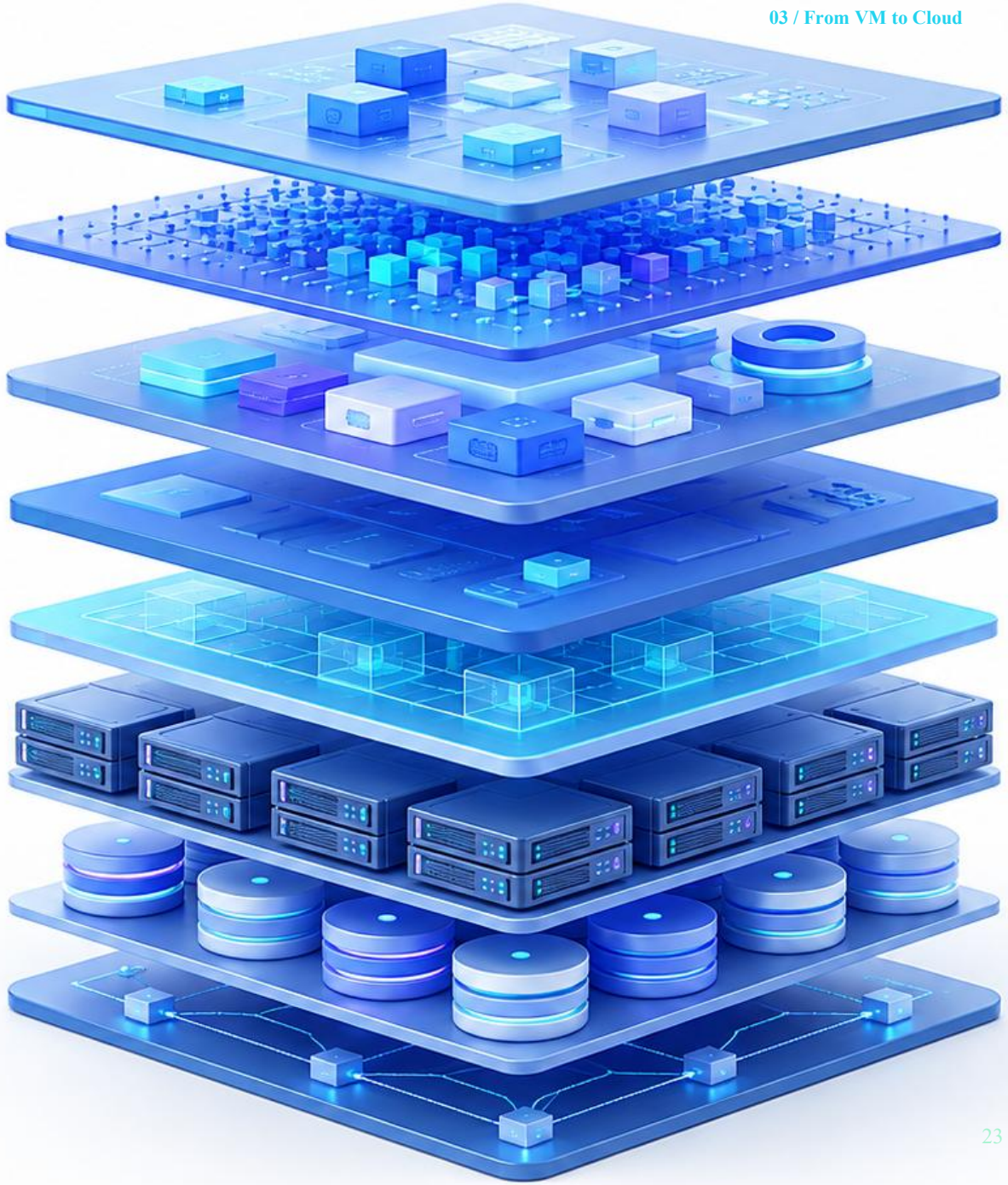
Limitations

- Performance overhead
- Resource contention
- Host failure impact
- Licensing concerns
- Management complexity
- Security misconfiguration risk

03

From VM to Cloud

Turning virtualized resources into network-delivered services



Virtualization ≠ Cloud Computing

They are related, but they answer different questions.

Virtualization

How can computing resources be abstracted from physical hardware?

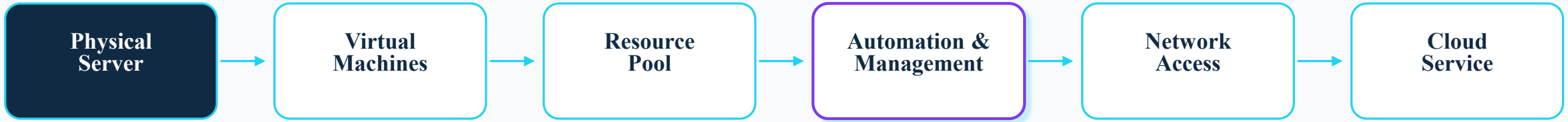
Cloud Computing

How can computing resources be delivered as services over a network?

Virtualization can enable cloud infrastructure, but not all virtualization is cloud.

From Server to Cloud

Cloud computing combines virtualization with automation, networking, management, and service delivery.



Cloud is not just “someone else’s VM.” It is an operating model for delivering resources on demand.

What Is Cloud Computing?

Computing resources are provided as services over a network.

Definition

Cloud computing provides resources such as compute, storage, platforms, and software through network-accessible services.

Plain-language view

Users request services; cloud infrastructure allocates and delivers the resources behind the scenes.



Core Characteristics of Cloud Computing

Cloud computing changes how resources are accessed, allocated, scaled, and measured.

On-Demand

Request resources when needed

Broad Network Access

Access services over a network

Resource Pooling

Share provider infrastructure

Rapid Elasticity

Scale capacity dynamically

Measured Service

Usage can be monitored and billed

Scalability vs Elasticity

Scalability expands capacity; elasticity adjusts capacity dynamically as demand changes.

Scalability

Planned increase or decrease in capacity as needs grow.

Example: moving from 2 servers to 10 servers.

Elasticity

Dynamic adjustment of resources in response to changing demand.

Example: automatically adding capacity during peak usage.

04

Cloud Service Models

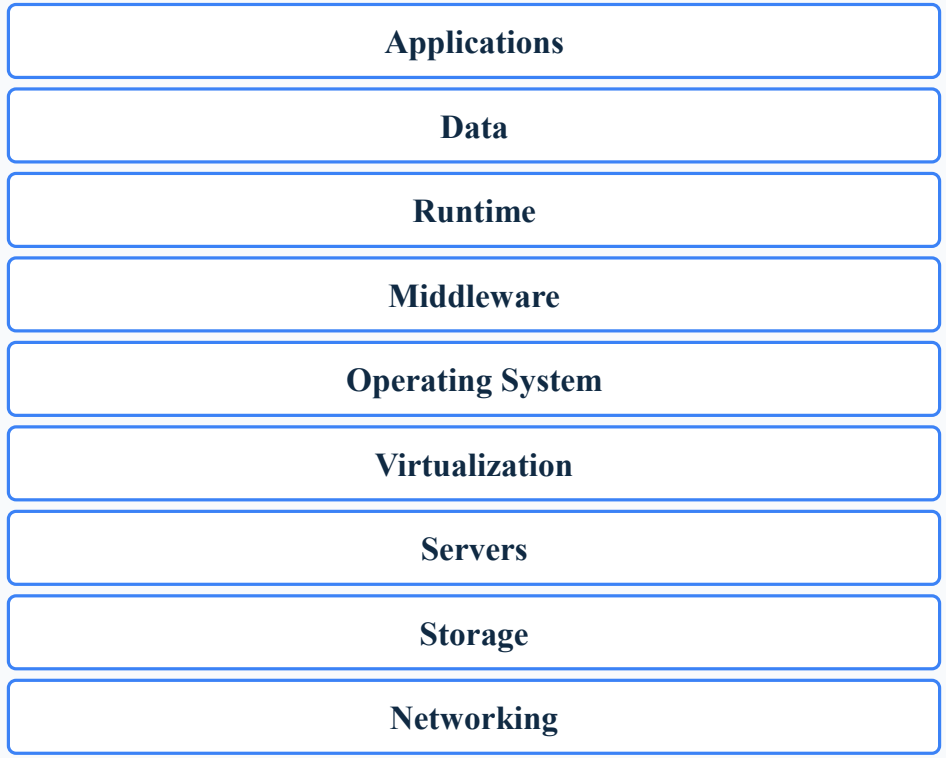
Understanding who manages each layer

The Traditional IT Stack

Before comparing cloud service models, identify the layers of responsibility.

Question

Who manages each layer?



Cloud models

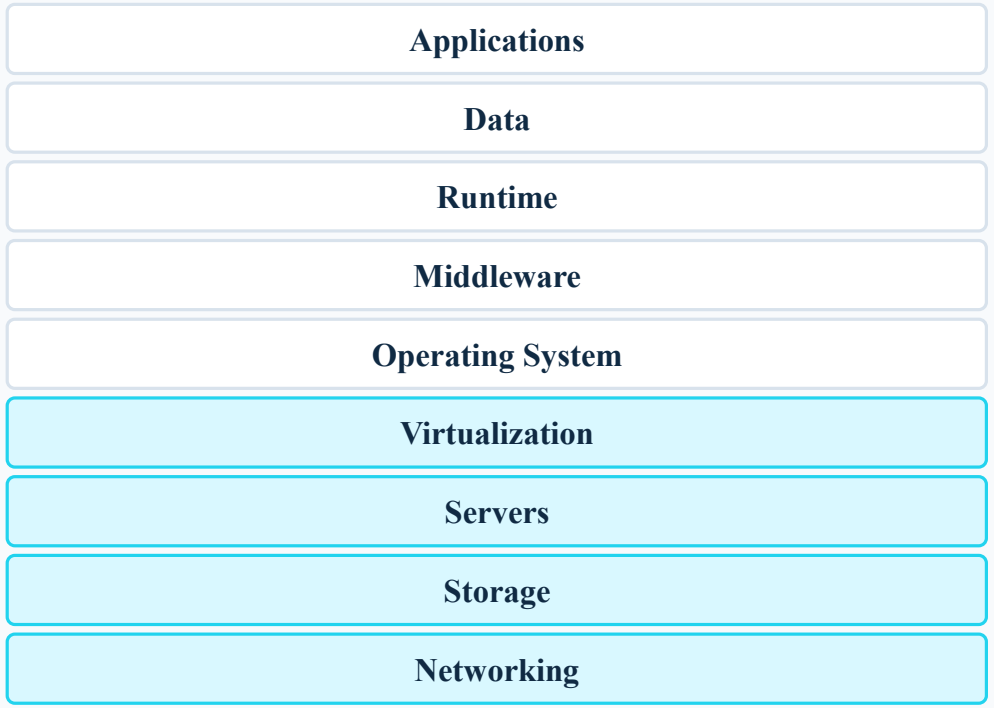
IaaS, PaaS and SaaS differ mainly by management responsibility.

IaaS — Infrastructure as a Service

Rent infrastructure; manage the software stack above it.

Customer manages

Outlined / white layers



Cloud provider manages

Filled / cyan layers

Key idea

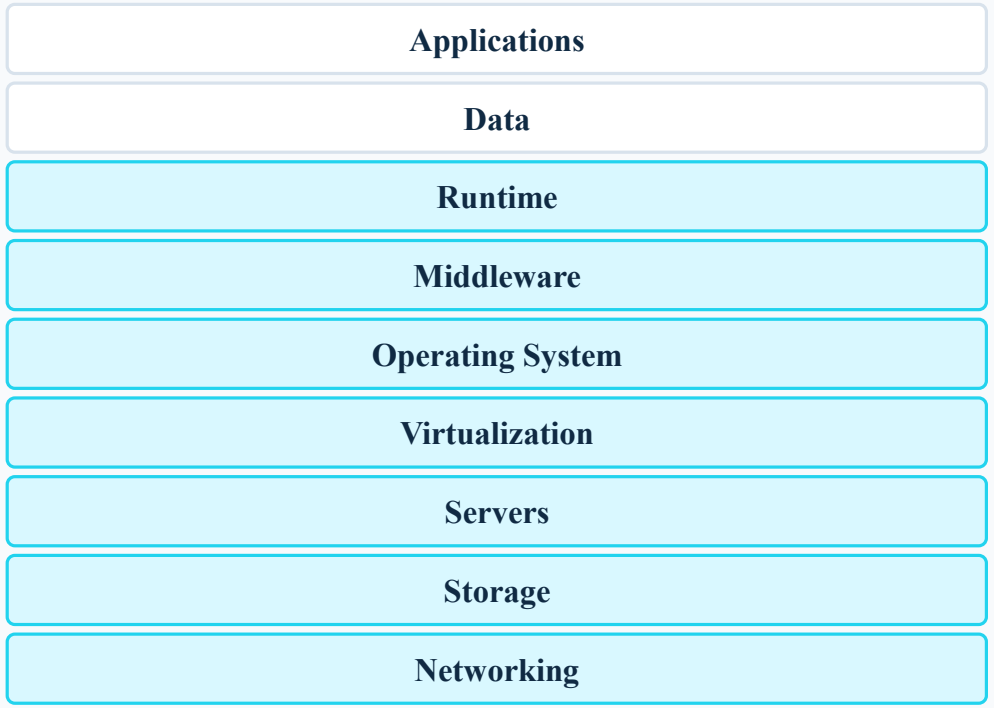
IaaS gives the customer high control over the OS and application layers while the provider manages the physical infrastructure.

PaaS — Platform as a Service

Build and deploy applications without managing most infrastructure.

Customer manages

Outlined / white layers



Cloud provider manages

Filled / cyan layers

Key idea

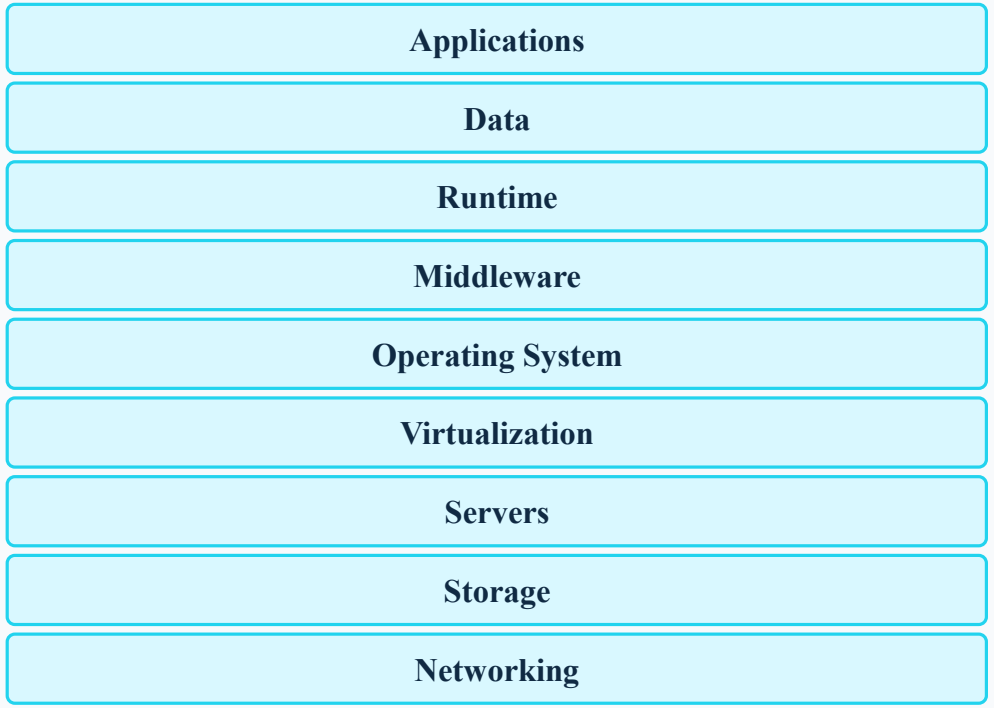
PaaS lets developers focus mainly on applications and data while the provider manages most platform layers.

SaaS — Software as a Service

Use a complete application delivered as a service.

Customer manages

Outlined / white layers



Cloud provider manages

Filled / cyan layers

Key idea

SaaS shifts most responsibility to the provider. Users consume software rather than manage the platform.

IaaS vs PaaS vs SaaS

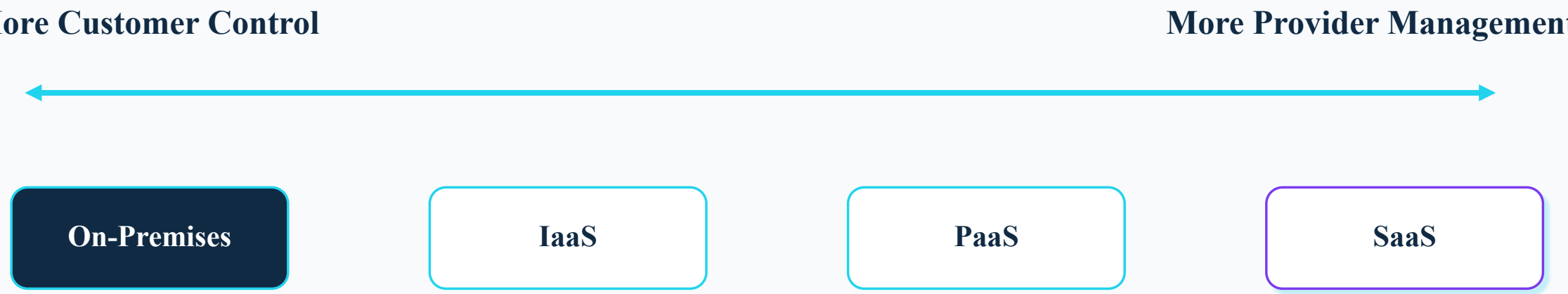
Service models differ by who manages each layer of the IT stack.

On-Premises	IaaS	PaaS	SaaS
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
OS	OS	OS	OS
Virtualization	Virtualization	Virtualization	Virtualization
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking

White = Customer manages | Cyan = Provider manages

Responsibility Gradient

Moving from on-premises to SaaS usually reduces customer management responsibility.



Memory cue

On-premises gives maximum control; SaaS gives maximum provider management.

05

Deployment & Practice

Choosing cloud environments and recognizing real-world trade-offs

Cloud Deployment Models

Deployment models describe where cloud resources live and how they are shared or connected.

Public Cloud

Shared provider infrastructure.

Private Cloud

Dedicated cloud environment for one organization.

Hybrid Cloud

Connected private and public environments.

Multi-Cloud

Use services from multiple cloud providers.

Hybrid Cloud vs Multi-Cloud

Hybrid and multi-cloud are related but not identical concepts.

Hybrid Cloud

Different infrastructure environments connected together.

Example: private infrastructure connected to public cloud resources.

Multi-Cloud

Use of services from multiple cloud providers.

Example: provider A for compute, provider B for analytics, provider C for backup.

Why Use Cloud Computing?

Cloud can increase flexibility, speed, access, and resilience when managed well.

CLOUD VALUE

- **Rapid deployment**
- **Flexible capacity**
- **Global access**
- **Managed services**
- **Backup / recovery**
- **Collaboration**
- **Automation**
- **Scalability**

Cloud is valuable when the service model matches the workload and the organization manages cost, security, and governance.

Cloud Risks and Challenges

Cloud computing changes operational risks rather than eliminating them.

- **Security**
- **Privacy**
- **Compliance**
- **Outages**
- **Internet dependency**
- **Vendor dependency**
- **Uncontrolled cost**
- **Configuration errors**

A good cloud design includes governance, monitoring, security configuration, and cost control.

Shared Responsibility for Security

Cloud provider and customer responsibilities differ by service model.

Cloud Provider
Security OF the Cloud

- Physical data center
- Core infrastructure
- Underlying platform
- Global service availability mechanisms

Customer
Security IN the Cloud

- User accounts
- Permissions
- Configurations
- Application data
- Access policies

Exact responsibility depends on IaaS, PaaS, and SaaS.

Containers vs Virtual Machines

Containers virtualize at the operating-system level; VMs virtualize hardware environments.

Virtual Machine

Application
Guest OS
Virtual Hardware
Hypervisor
Host Hardware

Stronger environment isolation, heavier footprint.

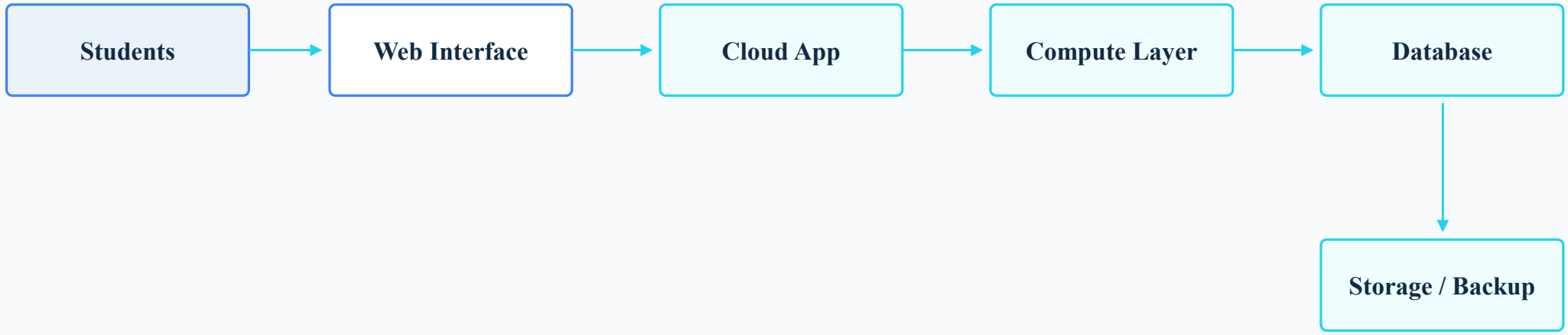
Container

Application
Dependencies
Shared Host OS
Container Runtime
Host Hardware

Lightweight packaging, common in modern cloud deployment.

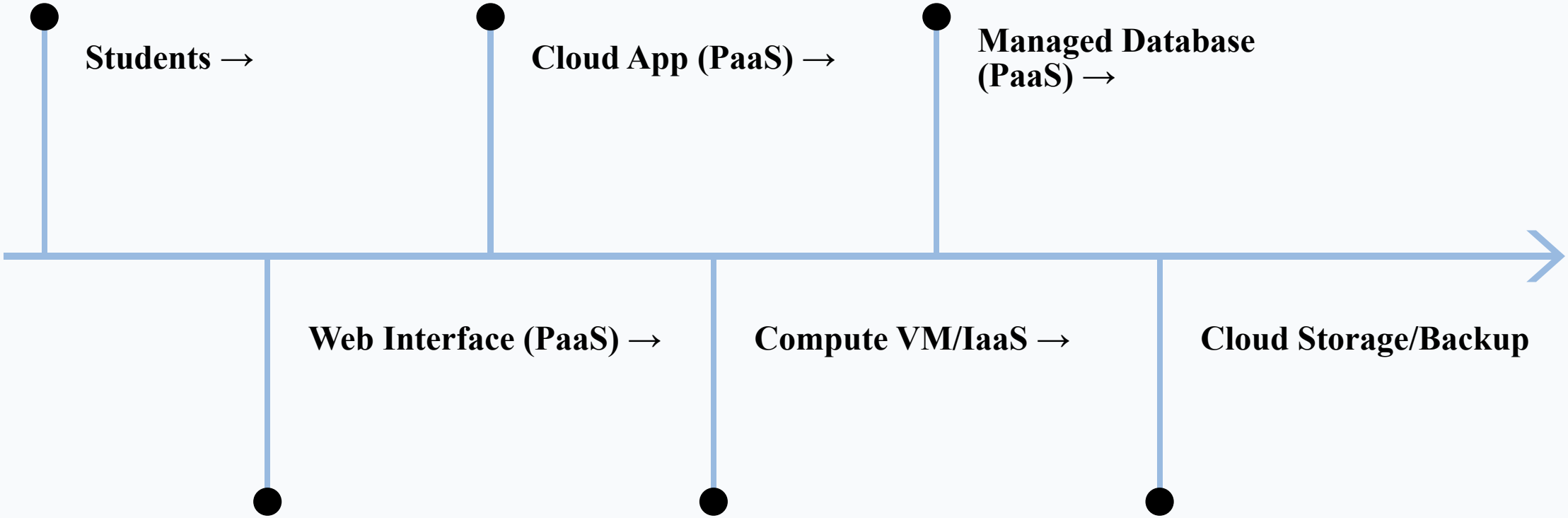
Scenario: University Registration System

Choose the right cloud approach by thinking about responsibility, scaling, and risk.



Discussion question

Which parts could be implemented as VM, IaaS, PaaS, or SaaS components?


Students →
Cloud App (PaaS) →
Managed Database (PaaS) →
Web Interface (PaaS) →
Compute VM/IaaS →
Cloud Storage/Backup

Part	Possible implementation	Why
Students	Not VM/IaaS/PaaS/SaaS	They are the end users accessing the service through devices and browsers.
Web Interface	PaaS	The university can deploy the web front end without managing the underlying servers and OS.
Cloud App	PaaS	Developers focus on application code and data while the provider manages runtime, OS, scaling, and infrastructure.
Compute Layer	VM / IaaS	Virtual machines can provide CPU, RAM, networking, and OS resources. A cloud VM is a common example of IaaS.
Database	PaaS / managed database	The provider can manage the database engine, patching, backups, and availability while the university manages its data.
Storage / Backup	IaaS or managed cloud storage	Cloud object/file storage provides infrastructure-level storage without buying physical storage hardware.

Classification Activity

Classify each scenario as VM, IaaS, PaaS, or SaaS.

Scenario 1

Run Linux inside your Windows laptop.

Scenario 2

Rent a virtual server online.

Scenario 3

Deploy code without managing the OS.

Scenario 4

Use an online email application.

Scenario 5

Test software in an isolated OS.

Scenario 6

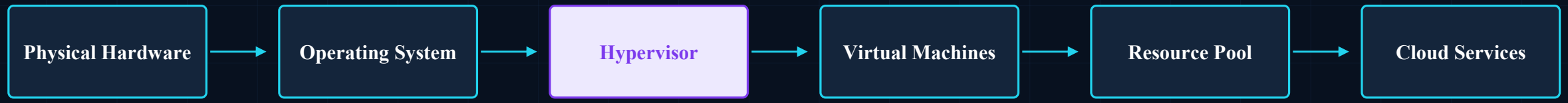
Use a cloud-based learning platform.

Students answer first, then discuss why responsibility and delivery model matter.

Scenario	Answer	Why
1. Run Linux inside your Windows laptop.	VM	Linux runs as a guest operating system inside a virtual machine on the Windows host.
2. Rent a virtual server online.	IaaS	The cloud provider supplies virtualized compute infrastructure, while you typically manage the OS and applications.
3. Deploy code without managing the OS.	PaaS	You focus on application code while the provider manages the OS, runtime, and underlying infrastructure.
4. Use an online email application.	SaaS	You use a complete software application delivered over the Internet without managing its infrastructure.
5. Test software in an isolated OS.	VM	A VM provides an isolated operating-system environment for testing.
6. Use a cloud-based learning platform.	SaaS	Students and instructors use a complete application delivered through the cloud.

From Hardware to Cloud Services

Key takeaway: virtualization abstracts resources; cloud computing delivers resources as services.



Takeaway 1

Virtualization abstracts physical resources.

Takeaway 2

A VM behaves like an independent computer.

Takeaway 3

IaaS, PaaS, and SaaS differ by management responsibility.

Exit question: If virtual machines already exist, why do we need cloud computing?